Mochamad Lutfi Firmansyah

mochamad.firmansyah@kfupm.edu.sa

ORCiD: 0000-0003-1844-0947

(+62) 87851129630

(+966) 597079742

EXPERTISE

Functionalized adsorbent

Metal recycling

Adsorption

Statistical Modelling

Complete Publication List

ACADEMIC QUALIFICATION

DOCTOR OF ENGINEERING

Kyushu University 2016-2019

- Solvent extraction, metal recycling, membrane transport

MASTER OF PHILOSOPHY

Universiti Teknologi Malaysia 2014-2015

- Mesoporous materials and fixed-bed catalysis

BACHELOR OF SCIENCE

Institut Teknologi Bandung 2009-2014

- Zeolite-based mesoporous materials

EMPLOYMENT HISTORY

POST-DOCTORAL FELLOW 2023-NOW

King Fahd University of Petroleum and Minerals

Interdisciplinary Research Center for Refining and Advanced Chemicals.

LECTURER 2019-2025

Universitas Airlangga

Nanotechnology Engineering
Faculty of Advance Technology and Multidiscipline

RESEARCH ASSISTANT 2015-2016

Universiti Teknologi Malaysia

Faculty of Chemical Engineering
Green Technology and Advanced Materials

LABORATORY ASSISTANT 2013-2014

Institut Teknologi Bandung

Faculty of Mathematics and Natural Sciences
Department of Chemistry
Inorganic Laboratory

BRIEF SUMMARY

Currently, I am a postdoctoral research fellow at King Fahd University of Petroleum and Minerals. I've worked and developed various adsorbents for wastewater treatment. I was a faculty member at a reputable university in Indonesia for over 3 years. I have experience as a teaching staff to a bachelor's degree student in a field related to chemistry and material engineering.

In addition to teaching and administrative work, my current research focus is developing a low-cost and highly functionalized adsorbent. Previously, our team delved into the urban mining landscape by developing e-waste recycling strategy in Indonesia, especially from mobile phone waste. I have also worked with several organization to aid in the development of their urban mining strategy.

INDUSTRIAL PROJECT

- **Banking Industry E-waste Technoanalysis study -** Bank BCA Tbk. 2023
- **Technology Options for the Purification of the Base Stream from the Esterification Column -** SIPCHEM. 2024
- **Development of Green VES-based Diverting Acid Fluids for Well Stimulation -** SIPCHEM (Proposal, 2024)

RESEARCH GRANT

- **Palm Research Grant -** Indonesian Palm Oil Plantation Fund Management Agency - 2022
- **Kurita Overseas Research Grant** - Kurita Water and Environment Foundation **-** 2023
- **Joint Research Scheme** - SATU President's Forum - 2022, 2023

TOP PUBLICATION

1. **M.L. Firmansyah,** I.N. Rizki, N. Ullah. Recent advances in urban mining technology: A focus on electronic waste recycling potential in Indonesia. Cleaner Waste Systems, 10, 100239 (2025
2. **M.L. Firmansyah,** M. Ashraf, N. Ullah. A critical review on the removal of anionic dyes by cross-linked resin: Recent progress, challenges and future perspective. Separation and Purification Technology, 360, 131111 (2025)
3. **M.L. Firmansyah,** N. Ullah. Ionic liquid-containing polymer including membrane for the removal of cobalt: Synthesis, adsorption, transport, and modeling studies. Separation and Purification Technology, 353, 128332 (2025)
4. **M.L. Firmansyah**, N.I. Wulan, A.P. Nurisca, V. Karthickeyan, A.A. Jalil. Palm Biomass-Derived Activated Carbon for the Recovery and Separation of Pt(IV) 1 from a Simulated Leachate. Biomass and Bioenergy, In press, (Januari 2024)
5. I.N. Rizki, I. Amalina, N.S. Hasan, N.F. Khusnun, A.A. Jalil, **M.L. Firmansyah**. Functionalized agriculture-derived biomass-based adsorbent for the continuous recovery of gold from a simulated mobile phone leachate. Chemosphere, 345, 140455 (December 2023)
6. **M.L. Firmansyah**, I. Amalina, I.N. Rizki, A.M. Alfaza, P.K. Jiwanti, A.A. Jalil, M. Goto. Functionalization of Biomass-derived Activated Carbon and Electrochemical Reduction for the Recovery of Gold from Mobile Phone Leachate, Arabian Journal for Science and Engineering, 1-13 (October 2023)
7. **M.L. Firmansyah**, W. Yoshida, T. Hanada, M. Goto. Application of Ionic Liquids in Solvent Extraction of Platinum Group Metals, *Solvent Extraction Research and Development, Japan*, 27, 1-24, (2020).
8. **M.L. Firmansyah**, F. Kubota, W. Yoshida, M. Goto, Application of a Novel Phosphonium-based Ionic Liquid to the Separation of Platinum Group Metals from Automobile Catalyst Leach Liquor, *Industrial & Engineering Chemistry Research*, 58, 3845-3852, (February, 2019).
9. **M.L. Firmansyah**, A.A. Jalil, S. Triwahyono, H. Hamdan, M.M. Salleh, W.F.W. Ahmad, G.T.M. Kadja. Synthesis and characterization of fibrous silica ZSM-5 for cumene hydrocracking. Catalysis Science & Technology. 6,(13), 5178-5182 (2016)

TEACHING EXPERIENCE

1. Basic Chemistry (2 Credit)
2. Spectroscopy (2 Credit)
3. Laboratory 1-4 (2 Credit)
4. Introduction to Nanotechnology (2 Credit)
5. Laboratory Management and Safety (2 Credit)
6. Material and Metallurgy (2 Credit)
7. Material Characterization Technique 1-2 (2 Credit)
8. Material Engineerign (3 Credit)

PUBLISHING EXPERIENCE

- Adsorption Science & Technology - Reviewer
- Arabian Journal for Science and Engineering - Reviewer
- Environment, Development and Sustainability - Reviewer
- Journal of Environmental Chemical Engineering - Reviewer
- Journal of Hazardous Materials - Reviewer
- Journal of Material Cycles and Waste Management - Reviewer
- Results in Engineering - Reviewer
- Cellulose - Reviewer
- Scientific Reports - Reviewer
- Malaysian Journal of Catalysis - Reviewer
- Atom Indonesia - Reviewer
- Alchemy - Reviewer
- Indonesian Journal of Chemistry - Reviewer

- Malaysian Journal of Catalysis - Associate Editor

CONFERENCE EXPERIENCE

- **Invited Speaker** - 3rd International Conference of Advanced Technology and Multidiscipline - 2023
- **Oral Presenter** - 18th International Conference on Clean Energy - 2022

BOOK

APA SIH NANOTEKNOLOGI ITU? (WHAT IS NANOTECHNOLOGY?)
ISBN: 9786024737641

HA? NANO!?
ISBN: 9786026635642

COMMUNITY PROJECT

AYO EKSPERIMEN
Real life science experiment with kids under 12 years old
https://www.youtube.com/channel/UC6dgtgO9EvnF83hL3saqYCw

REFERENCE

PROF. NISAR ULLAH

Department of Chemistry
King Fahd University of Petroleum and Minerals
nullah@kfupm.edu.sa

PROF. DR. AISHAH ABDUL JALIL

Chemical Engineering
Faculty of Chemical Engineering
Universiti Teknologi Malaysia
aishahaj@utm.my

PROF. MASAHIRO GOTO

Department of Applied Chemistry
Graduate School of Engineering
Kyushu University
m-goto@mail.cstm.kyushu-u.ac.jp